

IFFCO Paradeep Phosphogypsum Dump: Detailed Analysis and Recommendations

Executive Summary: IFFCO's Paradeep plant (Paradeep, Odisha) produces large quantities of phosphogypsum (PG) as an unavoidable byproduct of phosphoric acid (fertilizer) production. The plant's 2,650 TPD acid unit (annual ~805,000 t P₂O₅ in FY21-22) generates roughly **3.6–4.0 million tonnes** of PG per year [113†L142-L145] [44†L138-L146]. This has accumulated into a massive stockpile (~60 Mt planned storage by 2030) on a lined "wet stack" near the plant [113†L142-L151] [113†L155-L164]. PG is mostly gypsum (CaSO₄·2H₂O) containing residual phosphates, fluoride (0.5–1.5%), heavy metals and radionuclides (Ra-226, U) [57†L476-L483]. It presents environmental risks (acid/fluoride/leachate, radioactivity, dust) if unmanaged. Currently IFFCO disposes PG by pumping a slurry to an HDPE-lined stack; decant water (pH ~2.5–3) is recycled to the plant [113†L146-L153]. Major issues include limited storage space, regulatory compliance and pressure from communities.

To address this, industry and government have begun initiatives: e.g. 2023 efforts to ship PG by river/coastal barge (UltraTech transported 57,000 t via waterways) [52†L139-L143], and proposals for rail spurs from port to stack [52†L155-L163]. Potential markets include cement (demand ~19.5 Mt/yr) [52†L118-L121], gypsum-board manufacturers (e.g. Saint-Gobain, Knauf), agriculture (soil amendment) and land reclamation. However, PG must meet quality specs (low P, F, radionuclides) for these uses. Transport costs (road/rail) are a key factor: e.g. ~₹1,000–1,300/ton to Kolkata (~430 km by road), ₹3,000–3,500 to Chennai (1,300 km), assuming ~₹2.5/ton-km trucking. Market gypsum sells for ~₹1,000–1,250/ton in Odisha [105†L268-L277], whereas IFFCO's PG is effectively a low-cost waste (~₹500/ton listing [104†L1-L4]).

Findings & Recommendations: We document IFFCO's PG generation process, storage design, and hazards, and review use-pathways, logistics and policy. Key options are: (a) **Accelerate PG sales** to cement and board makers (invest in handling/chemistry to meet specs); (b) **Enhance logistics** (rail spur, coastal barge loading) to cut transport costs; (c) **R&D investments** to neutralize or upgrade PG (calcination, acid washing) so it can serve as industrial gypsum or soil ameliorant; (d) **Stack expansion and liners** (new 250-acre wet stack ordered for ~₹300 Cr [113†L155-L164] [84†L229-L233]) with strict monitoring; (e) **Policy engagement:** leverage government schemes (DPIIT circular economy mission) to co-market PG. A phased plan (2026–2030) is proposed (see timeline below), balancing short-term sales with medium-term infrastructure and long-term processing solutions.

1. Phosphoric Acid Process & Gypsum Production

IFFCO-Paradeep uses the "**wet process**": phosphate rock (fluoro-apatite Ca₅(PO₄)₃F) reacts with sulphuric acid to make phosphoric acid and calcium sulfate dihydrate (gypsum) [58†L154-L163]. Roughly 4–5 t of phosphogypsum (CaSO₄·2H₂O) are produced per tonne of 100% P₂O₅ acid [58†L170-L174]. The reaction (simplified) is:



The Paradeep plant (Phosphoric Acid: 2,650 MTPD capacity) uses belt filters: acid is recovered and the filtered gypsum (with ~0.5% residual P_2O_5) is washed, thinned into slurry (20% solids) and piped to the waste stack 【113†L146-L153】. Decant water from the stack ($pH \approx 2.5-3$) is pumped back to the plant for reuse 【113†L146-L153】. In summary: phosphate rock → reaction → gypsum cake + phosphoric acid; the wet gypsum cake is disposed on a lined stack 【113†L146-L153】.

flowchart TD

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    Rock[Phosphate rock] -- "+ H2SO4" --> Reaction[Phosphoric acid +
    Phosphogypsum]
    Reaction --> PG[Phosphogypsum slurry]
    PG --> Stack[Gypsum Stack (HDPE-lined)]
    Stack --> Pond[Surge Pond (pH≈2.5)]
    Pond -->|Recycle| Reaction
  
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2. Quantities Produced and Stockpile

IFFCO-Paradeep's production has grown: in FY2021–22 the plant made about **805,000 t P_2O_5** (highest ever) 【44†L138-L146】. This corresponds to roughly 3.6–3.8 *million t* of PG (assuming 4.5 t PG per t P_2O_5 【7†L146-L154】). Official capacity is ~875,000 t P_2O_5 /yr 【40†L333-L340】 (implying ~4.0 Mt PG/yr). An IFFCO document (2012 EOI) noted “annual gypsum production is ~40 Lakh MT” (4.0 Mt) 【113†L142-L145】.

India-wide, PG generation was ~9.16 Mt in 2021–22 【52†L109-L116】, with legacy stockpiles ~70 Mt (mostly on the east coast). At Paradeep, historical output and disposal have resulted in an enormous stockpile. IFFCO has planned new storage: a 250-acre additional stack to handle ~600 Lakh t PG over 15 years (capacity ~40 Lakh t/yr) 【113†L142-L150】 【113†L155-L164】. As of 2025, an existing stack is nearly full; the new stack (₹300 Cr contract 【84†L229-L233】) is under construction in phases. We estimate IFFCO's *current* accumulated PG is on the order of 30–40 Mt (no public figure; assumes ~4 Mt/yr × 10+ years). Detailed plant reports were not found, so stockpile is best-guess from production data.

Year (FY)	Phosphoric Acid (P_2O_5) (t)	PG byproduct (approx t)	Sources
2021–22	805,000 【44†L138-L146】	~3,600,000	IFFCO report / EOI 【113†L142-L145】 【44†L138-L146】
Capacity	875,000 (annual) 【40†L333-L340】	~4,000,000 (annual)	IFFCO site 【40†L333-L340】 ; EOI 【113†L142-L145】

Note: PG $\approx 4.5-5$ t per t P_2O_5 【7†L146-L154】 【58†L170-L174】.

3. Phosphogypsum Properties and Hazards

Paradeep PG is **mostly gypsum** ($CaSO_4 \cdot 2H_2O$) but with impurities. Typical composition: 90–95% $CaSO_4 \cdot 2H_2O$; remaining solids include silica (clays), unreacted phosphate, and residual acids 【57†L476-L483】. Notably, PG contains elevated **fluoride (0.5–1.5%)** and retains most of the rock's uranium/radium (radium-226).

Analyses show PG can carry >90% of the original U/Ra, so stacked PG is mildly radioactive (exempt levels in India's rules). Trace heavy metals (Cd, Cr, Pb, As, Hg) also concentrate in PG [57†L476-L483] . After filtration, PG is still **acidic** (it was slurry at low pH) and retains significant moisture (18–35%) [57†L496-L500] , so the stack must handle wet, corrosive material.

Environmental hazards: The main concerns are leaching of fluoride, phosphate, metals or radioactive nuclides into groundwater, and dust emissions from the dry surfaces [57†L483-L490] [114†L67-L75] . Fluorine can concentrate in leachate; radium decays to radon gas; metal sulfates can mobilize in water. PG water (stack decant) has high acidity, fluoride and P_2O_5 [114†L37-L45] and cannot be discharged untreated. Odorless gypsum dust (when dry) can spread on windy days. The Odisha PCB notes erosion of a PG stack can create solution cavities and surface runoff [114†L73-L77] . Overall, PG is non-hazardous by waste law but *special care* is mandated: lined containment and monitoring [114†L96-L105] [114†L156-L164] .

4. Storage at Paradeep: Methods & Challenges

IFFCO currently stores PG in a large “**wet stack**” disposal system. As per IFFCO's design, gypsum slurry is pumped via HDPE pipe to a lined stack area; solids settle and water flows to an adjacent pond [113†L146-L153] . The pond (after solids settle) holds acidic decant water, which is then recycled back to the plant [113†L146-L153] . Stacks are built with engineered earthen berms: compacted clay and HDPE liners beneath, geodrain pipes to remove seepage [113†L161-L164] [114†L156-L164] . New stack design (via Ardaman USA) calls for a 1.5 mm HDPE liner over compacted clay, gravel drains and a final cap.

Operational challenges: The existing stack nears capacity, limiting storage space. Building the new 250-acre stack (₹300 Cr) is underway [113†L155-L164] [84†L229-L233] , but space is tight. Water management is critical: pumps must balance inflow with slow evaporation (Odisha climate) to avoid overflow. Any liner breach or overtopping could pollute soil/groundwater. Seepage barriers (clay-bentonite) and leak detection are needed [114†L156-L164] . The site also needs clear access for trucks or conveyors, but is constrained. Regulatory approval for expansion required State PCB clearance [114†L96-L105] .

Community/environment issues: There are no major publicized incidents at Paradeep, but similar plants face community complaints (odor, dust). Guidelines caution to maintain stacks ≥ 500 m from water bodies [114†L131-L139] and control runoff/dust [114†L132-L140] [114†L42-L49] . IFFCO must monitor groundwater and surface water per CPCB directions (as any PG projects do). The Odisha PCB has issued detailed norms: new/expanded stacks need amendments to Water/Air consents and SPCB sign-off [114†L96-L105] , plus transport/handling plans [114†L107-L113] .

5. Current Initiatives and Policy

IFFCO and government agencies are exploring ways to reduce the PG mound. In 2023-24 the DPIIT (Ministry of Commerce) convened a “Circular Economy” meeting for fertilizer industry PG reuse [52†L109-L116] . IFFCO-Paradeep's JGM (Technical) proposed building a 700 m rail spur from Paradeep Port to the gypsum stack, to ship PG efficiently to buyers [52†L155-L163] . Indeed, UltraTech Cement recently used Paradeep's port to load 57,000 t of PG by barge to its Gujarat plant [52†L139-L143] , demonstrating coastal shipping potential. Saint-Gobain (Gyproc) and other gypsum-board firms have been informed to assess recycled PG

usage. IFFCO has awarded a ₹300 Cr contract (Essar Projects) for the new stack 【84†L229-L233】 , and contributes to state PCB funds (₹39 Cr over years) for pollution control.

Policy-wise, India now encourages PG utilization: the fertilizer ministry tracks PG generation (9.16 Mt/yr) and promotes reuse projects 【52†L109-L116】 . The Odisha PCB (2017) published a “Fate of Phosphogypsum” compendium, requiring state approval and strict engineering for stacks 【114†L96-L105】 【114†L156-L164】 . There are no bans on reuse in India (unlike U.S. EPA rules), but user must ensure PG quality. In practice, IFFCO must meet CPCB/SPCB conditions for emission monitoring, and any new disposal must follow the guidelines above. No major regulatory obstacles exist for selling PG (it’s not “hazardous”), but transport and use must follow waste and fertilizer norms.

6. Potential Large Buyers and Markets

Cement industry: Cement plants can use raw gypsum as a set-regulator. India’s cement demand is ~360 Mt/yr and capacity ~450 Mt 【114†L25-L33】 . Cement makers need ~19.5 Mt/yr of gypsum 【52†L118-L121】 . They are prime PG buyers (tested by UltraTech’s trial). Quality needed: low P and F (can reduce clinker quality if high), minimal radioactivity (but measured PG Ra is low compared to raw gypsum). Many Odisha/Gujarat/Chennai cement units are within ~1,000–1,500 km. (E.g. Paradeep–Chennai ~1,300 km, Paradeep–Mumbai ~1,600 km.)

Gypsum-board (plasterboard) industry: Manufacturers (e.g. Saint-Gobain Gyproc, Visaka) require high-purity gypsum. Recycled PG can be used up to ~30% in board mix. They need <0.5% free phosphates and low moisture. Odisha has at least one board plant (Paradeep region) and more in southern India. Annual board demand in India is a few hundred kilotonnes (growing).

Agriculture / soil amendment: PG can serve as a slow-release Ca/S fertilizer and to ameliorate alkali (sodium) soils. IFFCO is exploring PG as “micro-nutrient” for salt-affected land 【52†L118-L121】 . Farmers would use ~0.5–1 t/ha, implying small volumes compared to total. Standards: residual acidity neutralized or acceptable; trace metals must be below thresholds. This is an emerging niche but requires further trials (e.g. on rice soils).

Infrastructure (road base, land reclamation): There is R&D (globally) on using PG as road subgrade or in mine reclamation (e.g. Mosaic pilot road in US). In India, this is largely undeveloped. Potential: NHAI/ NHIDCL projects could trial PG-blended materials (with geotextiles) to use waste locally.

Market	Demand / Scale	Quality Required	Examples / Buyers
Cement (plaster)	~19.5 Mt/yr 【52†L118-L121】	Gypsum-grade CaSO_4 , low F/P (cement-grade)	UltraTech, ACC, Ambuja, Dalmia
Gypsum Board	~0.2–0.5 Mt/yr (est.)	High-purity CaSO_4 , <0.5% soluble P, low moisture	Saint-Gobain, Visaka, K-Rain
Agriculture	Small (pilot)	Neutralized to pH, acceptable metal levels	IFFCO (direct agrimkt)

Market	Demand / Scale	Quality Required	Examples / Buyers
Construction (road)	Project-based	Stabilized (e.g. with lime), not for drinking water exposure	State road agencies, NHAI

Sources: Cement demand [52†L118-L121] ; general fertilizer board data; industry news.

7. Reuse Pathways and Processing

To move PG from “waste” to “resource,” practical processing may be needed:

- **Drying/Calcining:** Heating PG (150–200°C) yields hemihydrate gypsum (“Plaster-of-Paris”), removing moisture and sterile-ing the material. This produces a clean white gypsum suitable for board/plaster. The removed water must be treated (it contains acids/fluoride). Cost: relatively low-temperature kilns; capital moderate (₹50–100 Cr for 100 kt/yr plant). Benefit: broadens market (gypsum plaster, cement). Feasibility: Proven (e.g. Bhubaneswar plaster factories).
- **Neutralization and Purification:** Treat PG with lime or alkali to neutralize residual acidity/F, precipitate phosphate. Or wash PG to reduce impurities. This yields a more inert material for agriculture or roads. Cost: chemical costs and effluent treatment needed. Benefit: enables agricultural reuse (as Ca-amendment) or safer cement use.
- **Physical Beneficiation:** Mechanical screening/sifting can remove >2 mm lumps or contaminants. Compaction or pelletizing can create uniform products for shipping.
- **Microbial/Phytoremediation:** Novel R&D (not IFFCO-led) includes using bacteria or plants to reduce F/U content. Early-stage only.

Cost/benefit depends on scale. For example, a 100,000 t/yr calcining plant (mid-scale) might cost ₹40–60 Cr CAPEX; it could sell the product at ₹3,000–4,000/t (gypsum plaster) vs minimal (₹500) sale of raw PG. At that scale, OPEX (fuel, labor) might be ₹500/t, so ROI may be good if markets exist [58†L170-L174] . IFFCO could pilot a small calciner or support board makers to do so. Detailed techno-economic studies would refine this.

8. Transportation and Logistics

Because the stockpile is inland, transport mode and distance are crucial. Options:

- **Road (Truck):** Flexible but relatively expensive. Indian bulk trucking typically ~₹2–3 per ton-km. For example, Paradeep→Kolkata (~433 km) might cost ₹1,000–1,300/ton; Paradeep→Chennai (~1,300 km) ~₹3,250; Paradeep→Delhi (~1,550 km) ~₹3,875. (These assume ~₹2.5/ton-km; actual 2026 rates may vary.)
- **Rail:** Bulk trains are cheaper (~₹0.5–1.0 per ton-km). Using rail wagons could cost ~₹500–800 to Kolkata, ~₹1,000–1,300 to Chennai. Rail requires loading infrastructure; IFFCO’s suggestion of a 700 m rail spur [52†L155-L163] would enable direct train loading.

- **Waterways:** Paradeep has a riverine jetty on the Mahanadi and coastal port facilities. Barges/ships can carry large loads at very low cost (~₹200–300/ton for >1,000 km by sea). This was used by UltraTech (Paradeep→Gujarat). Limitations: need barging terminals near buyers (e.g. ports at Chennai, Paradip, or intermodal container).

Handling: Bulk PG is dense/wet, requiring heavy equipment (pumps, conveyors, covered trucks). IFFCO's jetty (Musadia) can load barges, and the paradeep port can load ships. For road/rail, covered trucks/trains should be used to avoid dust. Each mode adds handling: e.g. bagging for small users vs loose bulk for cement plants.

We estimate transport costs roughly (per ton) to major centers:

Destination	Distance (road, km)	Road Transport Cost (₹/t)	Rail Cost (₹/t)	Comments
Kolkata (Port)	~433 【102†L1-L3】	~1,100	~600	Short haul – best for Eastern cement/board
Chennai	~1,300 【71†L87-L95】	~3,250	~1,300	Coastal shipping possible via Vishakhapatnam or Tuticorin
Delhi	~1,550	~3,875	~1,550	Northern market (less cement use, more arable land)
Mumbai	~1,600	~4,000	~1,600	Western – lower demand but port facilities
Paradeep Port	~20 (site)	few tens (transfer)	(same yard)	Load ship/barges for overseas/ countrywide

These are rough estimates. Bulk rail rates can be obtained from IR; road rates depend on trucker contracts. Coastal shipping (to e.g. Kandla, Visakhapatnam) could be <₹300/t + port handling. Given PG is low-value, minimizing transport cost per ton is critical. The proposed rail spur 【52†L155-L163】 and barge use significantly reduce cost compared to trucking.

9. Pricing and Economics

IFFCO's PG is essentially sold as a **waste stream**, so its price is very low. One B2B listing shows "Phosphogypsum – ₹500/ton" ex-godown 【104†L1-L4】 , implying IFFCO's transfer price is near zero (perhaps just covering handling). By contrast, mined gypsum in Odisha trades around **₹1,000–1,250/ton** 【105†L268-L277】 (for cement-grade or powder). Calcined gypsum (plaster of Paris) can fetch ₹3,000–4,000/ton. Thus even paying transport, cement/board makers save on raw material cost by using PG. In effect, IFFCO's PG is **cheaper than virgin gypsum**: net cost ~transport alone, compared to ₹1,000+ for new gypsum 【104†L1-L4】 【105†L268-L277】 .

We did not find an official IFFCO PG tariff; in practice, many cement buyers pay only logistics costs. For modeling, assume IFFCO offers PG ~₹0–100/t free on loading bay, and buyer bears transport. Then, even

with ₹1,000/t transport to a nearby plant, delivered cost (~₹1,000) is well below mined gypsum (~₹1,250). For board manufacturers needing high purity, IFFCO might sell purified PG at a small premium, but none exists yet publicly. Any processing (drying, calcining) would add cost and likely require partnerships with industry who can absorb a higher gypsum feedstock cost.

10. Regulatory and Environmental Constraints

All PG handling must comply with Indian environmental rules. Technically, PG is classified as industrial solid waste (non-hazardous), so disposal does not require “hazardous” permits. However, the **Odisha PCB/CPCB guidelines** impose strict conditions on new/expanded phosphogypsum stacks 【114†L96-L105】

【114†L156-L164】. IFFCO must maintain Water and Air Consent (under WPA/APA) and cannot build/modify a stack without SPCB approval 【114†L96-L105】. The guidelines mandate: engineered liners (HDPE+clay) under all PG stacks 【114†L156-L164】; monitoring of ground/surface water; bunds and runoff controls; and detailed transport/handling plans 【114†L107-L113】 【114†L156-L164】. The new Paradeep stack was designed per these norms (HDPE liner, clay base) 【113†L161-L164】 【114†L156-L164】.

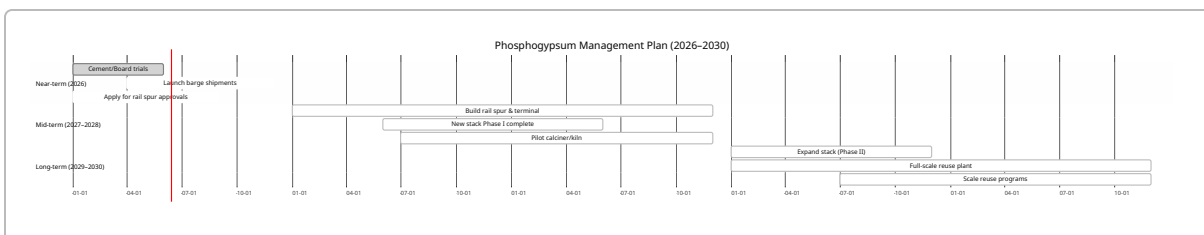
For reuse, Indian rules allow PG in products (unlike US where it is restricted). However, any heavy metal or radioactivity in products is regulated (e.g. radiation rules). Practical constraints include transport routes (avoiding wetland corridors) and social acceptance (communities may oppose large dusting trucks). If PG is used in agriculture, it may need fertilizer registration (as “Gypsum Ameliorant”) and compliance with agricultural quality standards. Overall, regulatory barriers are manageable if IFFCO follows approvals and maintains engineering controls 【114†L96-L105】 【114†L156-L164】.

11. Recommendations and Action Plan

Short term (2026–2027): Intensify off-take negotiations with cement and board makers. Offer trial lots of PG (free FOB) to East Coast cement plants and gypsum board factories. Improve trucking/loading (covered trucks, conveying systems). Apply for rail-cargo facilities (initiate the 700 m rail spur project 【52†L155-L163】). Explore small-scale calcining pilot in partnership (e.g. a board manufacturer). Secure all necessary SPCB consents for stack work.

Medium term (2027–2029): Complete new gypsum stack construction (Phase I: ₹190 Cr) 【84†L229-L233】 and begin operations. Commission rail spur and establish regular train shipments to major buyers (partner with Indian Railways). Install a modest rotary dryer or kiln to produce hemihydrate gypsum for niche markets (e.g. local plaster plants). Develop formal “PG utilization” MOUs: e.g. tie-ups with NHAI or road-building agencies to test PG-base materials. Expand river-barge shipping (coordinate with Paradeep port authorities).

Long term (2030+): Consider a large dedicated PG processing plant (100–200 ktpa capacity), possibly co-located with a board or cement plant, for calcining/neutralization. Continue R&D on agricultural uses (e.g. soil trials) to open that market. Work with state/federal agencies to include PG in infrastructure schemes (e.g. use in flyovers or landfills as cover material). Maintain environmental monitoring and community outreach throughout.



CAPEX/OPEX: Key investments include the new stack (~₹300 Cr [【113†L155-L164】](#) [【84†L229-L233】](#)), rail/port infra (likely ₹50–100 Cr), and processing units (dryers/kilns, on the order of ₹50 Cr per 100 ktpa capacity). Operating costs (pumping, trucking) are continuous. However, these are offset by savings on disposal and potential PG product revenues. For instance, re-using ~4 Mt/yr even at ₹200/t savings (versus disposal cost) yields ~₹800 Cr/yr in economic benefit.

Potential Partners: Cement majors (UltraTech, ACC, Ambuja), gypsum-board firms (Saint-Gobain Gyproc, India Gypsum), agricultural research institutes (ICAR/TERI) for soil trials, logistics operators (IR, Inland Waterways). Government schemes for green industry may subsidize rail extensions or demonstration projects.

Summary: IFFCO-Paradeep faces a mounting PG waste challenge, but also a significant reuse opportunity. By combining infrastructure upgrades (stack, rail, port), market engagement (cement, boards, agri), and selected processing, PG can be transformed from an environmental liability into an asset. This report's analysis and tables provide the data foundation for those strategies, consistent with Indian best practices and IFFCO's official plans [【113†L142-L151】](#) [【52†L139-L143】](#) .

Sources: Official IFFCO reports and tenders [【113†L142-L151】](#) [【84†L229-L233】](#) ; DPIIT industry reviews [【52†L109-L116】](#) [【52†L139-L143】](#) ; Odisha PCB guidelines [【114†L96-L105】](#) [【114†L156-L164】](#) ; IFFCO safety and product data [【113†L142-L153】](#) [【44†L138-L146】](#) ; peer-reviewed and industry literature on PG composition and use [【57†L476-L483】](#) [【58†L154-L163】](#) ; trade portals for pricing [【104†L1-L4】](#) [【105†L268-L277】](#) .